

1 Solution — GIAM 1.3

Problem 2

1.1 Problem

The sentence “Every metallic element is a solid at room temperature.” is false. Why?

1.2 The Short Answer

The sentence is **universally quantified** — it claims something about *every* metallic element. A universal statement is false as soon as there is a *single* exception, and there is one:

Mercury (Hg) is a metallic element that is *liquid* at room temperature.

Mercury melts at about -38.8°C , far below room temperature ($\approx 25^{\circ}\text{C}$), so at room temperature it is a *liquid*, not a solid. That one counterexample makes the statement false. ■

1.3 Why One Counterexample Suffices

The sentence has the logical form of a universally quantified statement — in fact a *universal conditional*:

$$\forall x (\text{Metal}(x) \rightarrow \text{SolidAtRT}(x)),$$

“for every element x , if x is a metal then x is solid at room temperature.” To *disprove* a “ $\forall$ ” statement, we use the **negation rule** for quantifiers:

$$\neg \forall x P(x) \equiv \exists x \neg P(x).$$

So the negation of our sentence is the *existential* statement

$$\exists x (\text{Metal}(x) \wedge \neg \text{SolidAtRT}(x)),$$

“there exists a metal that is *not* solid at room temperature.” To establish this existential, we need just **one** witness — and mercury is it. A single counterexample is logically

lete: it does not merely cast doubt on the universal claim, it *refutes* it outright.

1.4 The Counterexample, Visually

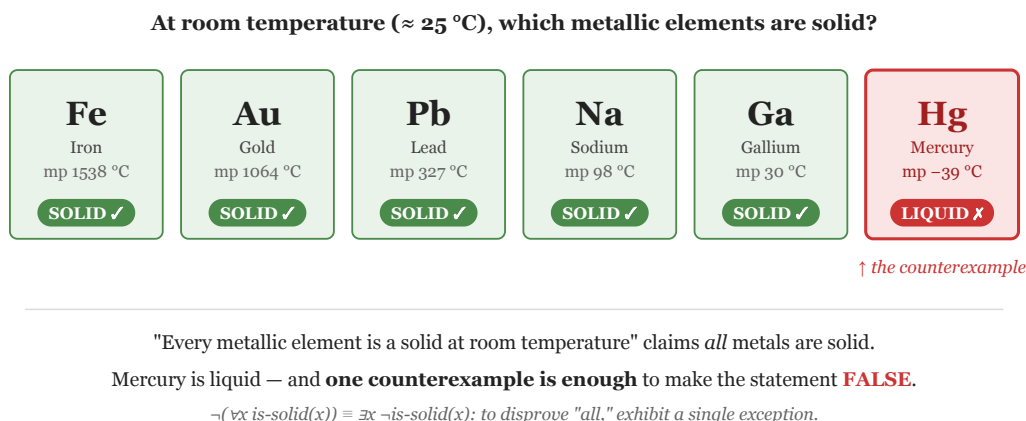


Figure 1.1: Iron, gold, lead, sodium, and gallium are all solid at room temperature; mercury alone is liquid (melting point -39°C). The single liquid metal is the counterexample that falsifies the universal claim.

1.5 Remark — Disproving “All” vs Proving “All”: An Asymmetry

There is a deep asymmetry between proving and disproving a universal statement, and this problem showcases the easy half:

- To **prove** $\forall x P(x)$ — that *every* metal is solid — you would have to verify it for *all* metals (infinitely many cases in general; here, every metallic element). No finite check suffices on its own; you need a general argument.
- To **disprove** $\forall x P(x)$ — to show it is false — you need only *one* counterexample.

This is why universal statements are often “easy to kill, hard to confirm.” We saw the same asymmetry in Chapter 1.2: the conjectures “ $2^p - 1$ is always prime” (Problem 5) and “ $x^2 - 31x + 257$ is always prime” (Problem 6) were each demolished by a single counterexample, while *proving* a universal prime-generating claim is generally hopeless. The mercury example is the chemistry-flavoured version of exactly that move.

1.6 Remark — Mercury Is the *Only* Liquid Metallic Element at Room Temperature

Mercury is special: among the metallic elements, it is the *unique* one that is liquid at standard room temperature. (The only other element that is liquid at room temperature is **bromine** — but bromine is a *nonmetal* (a halogen), so it is not a counterexample to *this* sentence, which is about metals.) Mercury’s liquidity is itself a striking physical fact, traceable to relativistic effects on its electrons that weaken the metallic bonding — which is why a single metal stands out as the exception.

1.7 Remark — The Definition of “Room Temperature” Matters

The sentence is false under *any* reasonable reading, but it is worth noting how close several other metals come — which is exactly why precise definitions (the recurring theme of these sections) matter:

Metal	Melting point	Solid at 25 °C?
Mercury (Hg)	−39 °C	no — liquid
Caesium (Cs)	28 °C	yes, but barely
Gallium (Ga)	30 °C	yes, but melts in your hand
Rubidium (Rb)	39 °C	yes
Sodium (Na)	98 °C	yes

Table 1.1

Caesium and gallium melt only a few degrees *above* a typical room, and **gallium famously melts in the warmth of a held hand (30 °C)**. If “room temperature” were taken to be a warm 30 °C day, caesium and gallium would *also* become counterexamples. But mercury is a counterexample regardless: at no ordinary room temperature is it solid. So the sentence fails robustly — mercury alone settles it, and the borderline metals only pile on.

1.8 Remark — How to State a *True* Version

It is instructive to ask what nearby statement *is* true. Several repairs work:

- **Weaken the quantifier** to existential: "*Some* metallic element is a solid at room temperature." — true (most of them are).
- **Restrict the universe**: "Every metallic element *other than mercury* is a solid at room temperature." — true under a 25°C reading (and very nearly true in general).
- **Change the predicate**: "Every metallic element is a solid *or a liquid* at room temperature." — true (no metal is gaseous at room temperature).

Each repair illustrates a standard logical move: when a universal claim is refuted by counterexamples, you either shrink the domain, weaken the quantifier, or loosen the predicate until the survivors all comply. Knowing *which* repair you want is the difference between a sloppy claim and a precise theorem — the discipline this section is training.